

NO. OF FUNCTIONAL FEATURES INCLUDED IN THE SOLUTION

Date	28 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Home Page	Displays project information and navigation menu	Frontend	Done	Medium
2	Prediction Form	Accepts weather parameters from the user	Frontend	Done	High
3	Input Validation	Validates user inputs before prediction	Frontend	Done	Medium
4	Data Preprocessing	Scales input data using StandardScaler	ML Module	Done	High
5	Flood Prediction	Predicts flood occurrence using XGBoost	Machine Learning Model	Done	High
6	Result Display	Displays flood prediction result	Flask Backend	Done	Medium
7	Model Loading	Loads trained model and scaler using Joblib	Backend	Done	High
8	Web Deployment	Deploys the application on Render	Deployment	Done	Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6
Features Tested & Verified	6
Additional / Nice-to-Have Features	4

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	Home Page, Prediction Form, Prediction Result
2	Backend / Logic	2	Flask Routing, Prediction Processing
3	Database / Storage	2	Saved XGBoost Model (.joblib), StandardScaler (.joblib)
4	API / Integration	0	Not Implemented
5	Security / Authentication	1	Input validation